NANYANG TECHNOLOGICAL UNIVERSITY SINGAPORE

SC2104/CE3002 Sensors & Interfacing
Tutorial 4 – Sensor Fusion

1. The general equation to implement the complementary filter to perform sensor fusion using the readings obtained from the Accelerator and Gyroscope is as follows:

$$\Theta[n] = (1- \alpha) * (\Theta_A[n]) + \alpha * (\Theta[n-1] + \omega_G[n] * \Delta t)$$

where

- Θ can be the Pitch or Roll angles derived based on sensor fusion
- Θ_A is the corresponding angle derived from the Accelerator's outputs values
- ω_G is the rate of change of the corresponding angle output by the gyroscope.

and α is the weightage associated with the Accelerator value and Gyroscope values.

Explain how the equation implements the complementary filter used to perform the sensor fusion function.